

Reading a Behavioral Code Across Mice, Days, and Model Classes: Longitudinal Leave-One-Animal-Out Decoding of Mesoscale Widefield Calcium Imaging

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Abstract

Decoding behavior from cortex-wide widefield calcium imaging is now routine, but almost always within a single animal. Whether a cortex-wide behavioral decoder transfers to an unseen individual—and, if so, whether that transfer is stable over days and robust to model capacity—has not been characterized. We study this on the open BraiDyn-BC cohort (25 mice, a 15-day cued lever-pull protocol; dorsal-cortex $\Delta F/F$ parcellated into 44 Allen-atlas regions), decoding four discrete task events (lever-pull, lick, reward, tone) with a strict leave-one-mouse-out (LOMO) protocol in the shared parcel space. A simple, interpretable linear decoder trained on $N-1$ mice reads every held-out mouse above chance for every event (LOMO AUC 0.80–0.93, permutation $p = 0.007$), landing within 0.03–0.04 AUC of the within-animal ceiling—a small residual cross-subject gap. Our contributions are what this gap does, not merely that it is small; we characterize it along two axes on the two highest-count events (lever-pull and lick). (i) Longitudinal stability: exploiting the 15-day protocol, a decoder trained on other mice’s first-week sessions reads a held-out mouse’s last-week sessions with no additional loss, and within-mouse accuracy is flat in the train–test day-gap (< 0.04 AUC over two weeks)—to our knowledge the first longitudinal characterization of a cross-animal cortical decoder. (ii) Capacity-invariance: nonlinear temporal models (a 1-D CNN and a GRU over the evoked window) raise cross-animal AUC by +0.09 to +0.12 while the cross-subject gap stays within a narrow 0.013–0.047 band and does not systematically widen with capacity, so the shared structure includes the response dynamics, not only the linear evoked mean. (iii) Replication and interpretability: the effect reproduces on an independent widefield cohort (different lab, atlas, and task), and decoder weights localize to a shared midline retrosplenial/visual-association backbone with target-specific somatosensory and motor modulation. We position these results relative to recently published atlas-aligned foundation-model work, which they complement with an interpretable, near-ceiling, longitudinally-controlled baseline. Code and a streamed-feature pipeline (no raw-video download) are released.

1 INTRODUCTION

Mesoscale (widefield) calcium imaging records activity across the dorsal cortical surface during behavior, and cortex-wide dynamics are strongly modulated by movement and task events (Musall et al., 2019; MacDowell & Buschman, 2020). Decoding behavior from this signal is now standard practice, but the overwhelming majority of studies train and evaluate within a single animal: a model is fit on some trials of one mouse and tested on other trials of the same mouse. This answers “can we read this brain?” It does not answer whether the cortex-wide behavioral code is the same across individuals—a question that determines whether a shared, atlas-referenced representation exists or whether each brain is idiosyncratic—nor whether any such shared readout is stable over days or robust to the decoder’s capacity.

Two facts make the cross-animal question non-trivial. First, vasculature, hemodynamic coupling, and atlas registration differ between mice, and these nuisances could dominate the parcellated signal. Second, even if a shared code exists at one moment, cortical representations are known to drift over days at the single-neuron level (Rule et al., 2020), so a decoder that transfers across mice on day 1 need not transfer on day 15. The recently released BraiDyn-BC resource (Kondo et al., 2025) is unusually well

suited to settle both: it provides cortex-wide $\Delta F/F$ parcellated into a shared 44-region Allen atlas from 25 mice, each recorded across a ~ 15 -session, multi-week operant protocol. Its data descriptor performs no decoding of any kind, leaving these questions open on the cohort.

We give a focused, interpretable answer. Working entirely in the shared 44-parcel space—so that “parcel j ” means the same anatomical region in every mouse and no per-subject alignment is needed—we perform strict leave-one-mouse-out (LOMO) decoding of four discrete task events, and then ask what happens to cross-animal readout across time and across model capacity.

Contributions.

- An interpretable near-ceiling cross-subject baseline. A 44-dimensional linear LOMO decoder reads every held-out mouse above chance for all four events (LOMO AUC 0.80–0.93, $p = 0.007$), within 0.03–0.04 AUC of the within-animal ceiling. This is a small, honestly reported residual gap—not a claim of zero gap.
- Longitudinal stability (primary result). Using the 15-day protocol, we show the cross-animal decoder does not degrade over time: trained on other mice’s early sessions (days 1–5), it reads a held-out mouse’s late sessions (days 11–15) with no additional loss, and within-mouse accuracy is essentially flat in the train–test day-gap. We are not aware of a prior longitudinal characterization of a cross-animal cortical decoder.
- Capacity-invariance. Nonlinear temporal decoders (1-D CNN, GRU) improve cross-animal AUC substantially without systematically widening the cross-subject gap, indicating the shared component of the code encompasses the temporal response and not merely its collapsed mean. (Contributions (i) and (ii) are established on lever-pull and lick, the two events with enough per-block trials to support the split.)
- Replication + interpretability. The pattern reproduces on an independent widefield cohort, and the decoders localize to interpretable, partly target-specific cortex.

We are explicit (Section 2) that cross-animal conservation of cortex-wide dynamics is already established and that a concurrent foundation-model preprint performs atlas-aligned cross-subject decoding on this dataset; our aim is the complementary, interpretable, and longitudinally-controlled characterization those works do not provide.

2 RELATED WORK

Cross-subject widefield modeling. Most directly related, WiCAT (Hosseini et al., 2026) learns atlas-aligned, subject-invariant representations of widefield imaging and performs zero-shot cross-subject behavior decoding on the same Kondo/BraiDyn-BC cohort (25 mice, 352 sessions), alongside a second (Musall) cohort, using self-supervised pretraining over spatiotemporal tokens registered to the Allen CCF without session-specific components. It establishes that cross-subject widefield decoding is achievable, and we accordingly claim no priority on that concept. Our contribution is complementary along three concrete axes. (a) Representation. We decode in the interpretable 44-parcel atlas space with a simple linear/CNN/GRU LOMO protocol rather than a learned-token foundation model, so the fitted weights are directly readable as cortical anatomy (Section 4.4). (b) Target and metric. WiCAT decodes continuous behavior on held-out subjects (reported zero-shot Kondo $R^2 = 0.18$ under linear probing, 0.29 with a Kalman decoder), whereas we decode discrete task events and report AUC. These quantities measure different prediction problems under different metrics and are not directly comparable; we make no claim of outperforming WiCAT. Read together they are nonetheless suggestive—near-ceiling cross-animal AUC on discrete events alongside modest zero-shot R^2 on continuous trajectories hints that the individual-invariant component of the code carries event structure more readily than moment-to-moment behavioral detail—though because architecture, supervision, and metric all differ simultaneously, we offer this as a conjecture for future controlled work rather than a result. (c) Uncharacterized axes.

Neither longitudinal stability nor capacity-invariance of cross-subject transfer is examined there; those are our contributions (i) and (ii).

Conservation of cortex-wide dynamics. That cortex-wide activity is shared across individuals is established. MacDowell & Buschman (2020); MacDowell et al. (2024) show a small set of cortex-wide spatiotemporal “motifs” explains $> 75\%$ of variance and generalizes across mice, with animals differing mainly in how often shared motifs are expressed; Musall et al. (2019) show cortex-wide single-trial activity is dominated by movement. These are unsupervised decompositions and within-animal encoding analyses; they motivate our premise but do not decode behavior across animals. Our conservation-adjacent result (the small LOMO-vs-within gap) is thus a confirmation of this consensus, not a discovery—which is why we foreground stability and capacity instead.

Cross-subject decoding in other modalities. Training a decoder on one animal and testing on another is established in electrophysiology via manifold alignment in rat sensorimotor cortex (Melbaum et al., 2022) and via contrastive embeddings that expose a “universal” hippocampal code (Schneider et al., 2023). These use different modalities/species and typically alignment-based matching rather than an alignment-free, shared-atlas LOMO decoder, and none address multi-day stability of the transfer.

Deep decoders on widefield imaging. End-to-end CNN–RNN models decode behavioral state from cortex-wide calcium imaging and outperform linear baselines (Ajioka et al., 2024), but for within-animal splits, pixel-level input, and locomotion rather than discrete events. Thus that nonlinear beats linear is known within animals; whether capacity changes cross-subject transfer—our question (ii)—is open.

3 DATA AND METHODS

Dataset. BraiDyn-BC (Kondo et al., 2025) (DANDI:001425, CC-BY) provides, per mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework regions (22/hemisphere) at ~ 30 Hz, with time-aligned behavioral channels (lever state, lick, reward, tone). Twenty-five mice each performed ~ 15 daily operant sessions spanning ~ 3 weeks. Raw movies are large (~ 5 GB/file) but the parcellated traces are tiny, so we stream only the 44-region $\Delta F/F$ and behavioral channels from cloud storage, never downloading raw imaging.

Events and features. For each target we detect event onsets as rising edges (1 s refractory). On the operant task sessions all four channels are well populated (reward fires on each rewarded trial, tone on each cue), so all four are decoded. For each onset we form a 44-dimensional evoked feature: mean post-onset (0–1 s) parcellated $\Delta F/F$ minus a pre-onset (−0.5–0 s) baseline. Negative examples are matched-count windows drawn > 2 s from any event. The cross-animal analysis pools three task sessions per mouse spread across the protocol.

Decoding and evaluation. The core decoder is a standardized ℓ_2 -regularized logistic regression. The evaluation axis is leave-one-mouse-out (LOMO): train on all events of $N-1$ mice, test on the held-out mouse; the 44 atlas parcels are shared across mice, so the split needs no per-subject alignment. We report mean held-out-mouse AUC, a cluster (over-mice) bootstrap 95% CI, and a permutation null in which event/baseline labels are shuffled within each mouse before re-running the full LOMO pipeline (150 permutations). As a positive control we run within-mouse 5-fold cross-validation. Per-parcel importance is the magnitude of the standardized logistic weights.

Longitudinal design. To probe temporal stability we pool each mouse’s daily task sessions into an early block (days 1–5) and a late block (days 11–15) and cross the within/across-animal axis with the same/across-block axis, giving four regimes: within-mouse same-block (5-fold CV on early; WS), within-mouse across-block (train early $\rightarrow$ test late, same mouse; WX), LOMO same-block (train other mice’s early $\rightarrow$ test held-out early; LS), and LOMO across-block (train other mice’s early $\rightarrow$ test held-out mouse’s late; LX). The across-block change is tested by a sign-flip permutation test on the per-mouse paired differences. A within-mouse train-day/test-day sweep over all ordered session pairs yields the accuracy-vs-day-gap curve and its slope.

Table 1: Cross-mouse (leave-one-mouse-out) decoding of four task events. Every held-out mouse is above chance for every target; the cross-animal AUC sits 0.03–0.04 below the within-mouse ceiling.

Target	Mice	Within-mouse AUC	LOMO AUC (95% CI)	p
Lever-pull	25	0.842	0.804 [0.780, 0.829]	0.007
Lick	25	0.885	0.850 [0.822, 0.874]	0.007
Reward	25	0.953	0.926 [0.911, 0.938]	0.007
Tone	25	0.864	0.828 [0.797, 0.859]	0.007

Nonlinear temporal decoders. To test whether transfer depends on model capacity we retain the full spatiotemporal window per event (-0.5 to $+1.0$ s, 45×44) and compare the linear evoked-vector model against a 1-D CNN over time (two Conv1d–BN–ReLU blocks, global average pool, dropout, linear head) and a single-layer GRU over the frame sequence, trained with Adam (BCE loss, weight decay, per-channel train-set standardization) under the same within-mouse and LOMO splits.

4 RESULTS

4.1 An interpretable cross-subject baseline: a small residual gap

All four task events decode from the cortex-wide response of a never-seen mouse well above chance (Table 1, Fig. 1): LOMO AUC ranges 0.804 (lever-pull) to 0.926 (reward), all at permutation $p = 0.007$ (the resolution floor of 150 permutations), with bootstrap lower bounds ≥ 0.78 . Every one of the 25 held-out mice exceeds chance for every target, with no per-mouse AUC below 0.65 (minima 0.653 lever-pull, 0.661 lick, 0.811 reward, 0.697 tone; Fig. 4, bottom)—these are point estimates, as we do not run a per-mouse significance test. Cross-animal decoding lands within 0.03–0.04 AUC of the within-animal ceiling for all four events (e.g. lever-pull 0.804 vs. 0.842; reward 0.926 vs. 0.953). Because the pooled cross-animal decoder is estimated from $\sim 24\times$ more data than any single-mouse model, and the within-mouse controls here are themselves well powered (three sessions/mouse), this small gap does not reflect a data imbalance: the behavioral code is largely shared across individuals in the atlas space. We treat this as an interpretable baseline, not the headline; the substantive results are how the gap behaves over time and capacity.

4.2 Longitudinal stability of the cross-animal decoder

Does the shared readout survive the passage of days? Using the four-regime design (Table 2, Fig. 2), we find it does. First, there is no detectable within-animal drift: training on days 1–5 and testing on days 11–15 of the same mouse costs nothing measurable—lever-pull WX 0.838 vs. WS 0.826 (a slight gain) and lick 0.865 vs. 0.883 (a 0.018 dip); the two directions disagree and both sets of CIs overlap, so we read this as no reliable change rather than as evidence of drift. Second, and centrally, the cross-animal decoder is temporally stable: trained only on other mice’s first-week sessions and tested on a held-out mouse’s last-week sessions (LX), it equals or exceeds the same-block cross-animal control (LS)—lever-pull 0.818 vs. 0.792 (paired $\Delta = +0.026$, sign-flip $p = 0.008$, $n = 25$) and lick 0.842 vs. 0.832 ($\Delta = +0.010$, $p = 0.44$). In neither cross-animal comparison is the across-block decoder worse; the mildly positive direction is consistent with late-week behavior being more practiced and thus slightly cleaner, not with temporal degradation. Finally, a within-mouse train-day/test-day sweep (> 2300 session pairs per target) shows accuracy is flat in the day-gap: slopes are -0.0004 (lever-pull) and -0.0022 (lick) AUC/day, i.e. < 0.04 AUC lost across the entire two-week span (Fig. 2, right). The cross-animal behavioral code is thus stable along both the individual and the session axes.

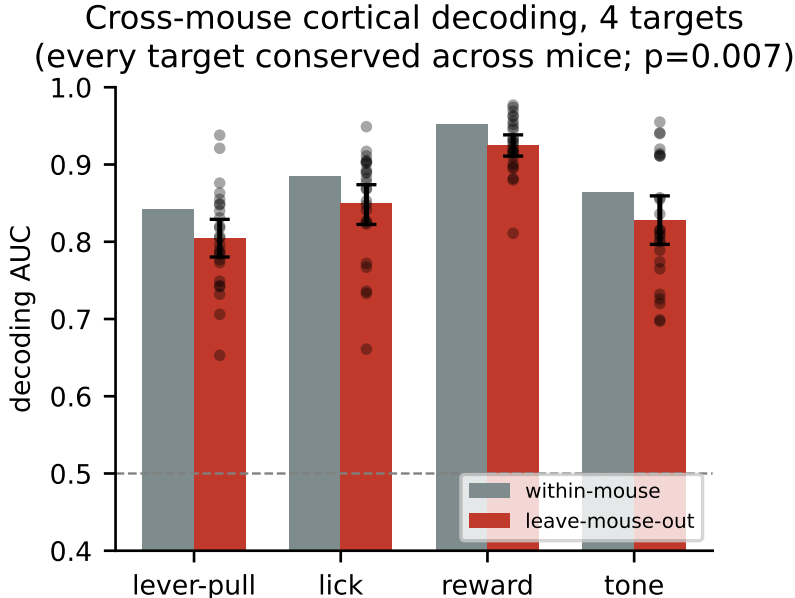


Figure 1: Within-mouse vs. leave-one-mouse-out AUC for all four targets (bootstrap CI; dots = held-out mice). Cross-animal decoding approaches the within-animal ceiling—a small residual gap.

Table 2: Longitudinal design: within/across-animal $\times$ same/across-block AUC. Training on days 1–5 and testing on days 11–15—within a mouse or across mice—incurs no loss.

Target	within-mouse		leave-one-mouse-out	
	same (WS)	early→late (WX)	same (LS)	early→late (LX)
Lever-pull	0.826	0.838	0.792	0.818
Lick	0.883	0.865	0.832	0.842

4.3 Capacity-invariance: the gap does not grow with model capacity

The linear baseline collapses each event to one evoked-difference vector, discarding temporal dynamics. A higher-capacity model could exploit those dynamics—but it could also overfit mouse-specific idiosyncrasies and open a cross-animal gap. We test both by comparing the linear model to a 1-D CNN and a GRU over the full window on identical events and splits (Table 3, Fig. 3). Both temporal models raise absolute accuracy sharply: cross-animal AUC rises from 0.804 to 0.895/0.909 (CNN/GRU) for lever-pull and from 0.850 to 0.966/0.962 for lick—a +0.09 to +0.12 gain, so the collapsed vector leaves real signal in the dynamics unused. Crucially, the cross-subject gap does not grow systematically with capacity: across all six model $\times$ target combinations it stays within a narrow 0.013–0.047 AUC band, and in three of the four nonlinear-vs-linear comparisons it is smaller than the linear model’s (lick CNN -0.013 and GRU -0.013 vs. linear -0.036 ; lever-pull GRU -0.026 vs. linear -0.037). The single exception is the lever-pull CNN (-0.047 vs. linear -0.037); we report this 0.010 excess for completeness but do not interpret it, as we run no test comparing one model’s gap against another’s. The shared component of the code therefore includes the temporal response and not merely its collapsed mean: the temporal models buy +0.09–0.12 AUC without a systematic cost in transfer.

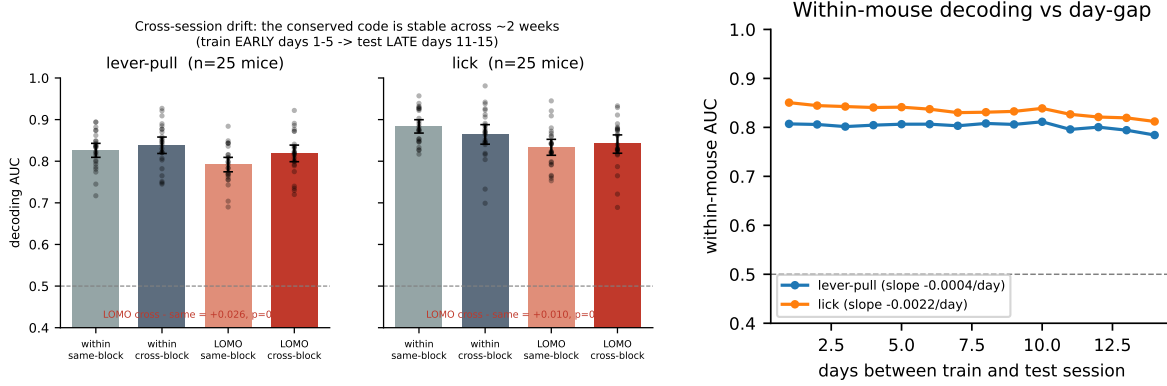


Figure 2: Left: within/across-animal $\times$ same/across-block decoding (bootstrap CI; dots = mice); the conserved code is temporally stable. Right: within-mouse accuracy vs. train–test day-gap is flat (slopes -0.0004 and -0.0022 AUC/day).

Table 3: Linear vs. nonlinear temporal decoders. Temporal models are far more accurate, yet the cross-subject gap (LOMO – within) stays within 0.013–0.047 and does not systematically exceed the linear model’s (smaller in three of four comparisons; larger only for the lever-pull CNN, by 0.010). Within-mouse and LOMO values are separate runs from the linear entries of Table 1, so they can differ in the third decimal.

Target	Model	Within-mouse	LOMO	Gap
Lever-pull	Linear (evoked)	0.841	0.804	−0.037
	1-D CNN (temporal)	0.942	0.895	−0.047
	GRU (temporal)	0.935	0.909	−0.026
Lick	Linear (evoked)	0.886	0.850	−0.036
	1-D CNN (temporal)	0.978	0.966	−0.013
	GRU (temporal)	0.975	0.962	−0.013

4.4 Cross-dataset replication and cortical localization

Replication. To test dataset-specificity we repeat the analysis on an independent widefield cohort from a different laboratory (Cardin–Higley; Benisty et al., 2024; 6 mice, 23 CCFv3 parcels, spontaneous behavior, different targets). Movement (locomotion) reaches LOMO AUC 0.731 (within-mouse 0.744) and arousal (pupil) 0.627 (within 0.645): the same near-ceiling cross-animal readout (gap 0.01–0.02) appears across a different lab, atlas, task, and behavioral variable, indicating a general property of cortex rather than a cohort artifact.

Localization. Decoder weights (Fig. 4) are spatially structured. Ranking parcels by standardized-weight magnitude, exactly three of the 44 appear in the top eight of all four targets: retrosplenial agranular cortex (RSPagl, left) and anteromedial visual-association cortex (VISam, bilaterally)—a compact midline retrosplenial/visual-association backbone, as expected when a movement/arousal-related component pervades widefield activity (Musall et al., 2019). Target identity modulates the remainder: the tone (auditory-cue) decoder uniquely recruits nose and upper-limb somatosensory cortex (SSp_n, SSp_ul) together with motor cortex (MOp); the lick decoder recruits lower-limb somatosensory cortex bilaterally (SSp_ll); and the reward decoder recruits neither, its top parcels being the shared backbone plus posterior parietal (VISa), dorsal retrosplenial (RSPd), and barrel cortex (SSp_bfd). The codes thus carry target-specific structure over a shared backbone rather than being pure generic arousal—though we note the somatosensory subfields that separate the targets are not the ones a naive somatotopy of each action would predict, so we report the dissociation without interpreting its anatomy. Parcel names

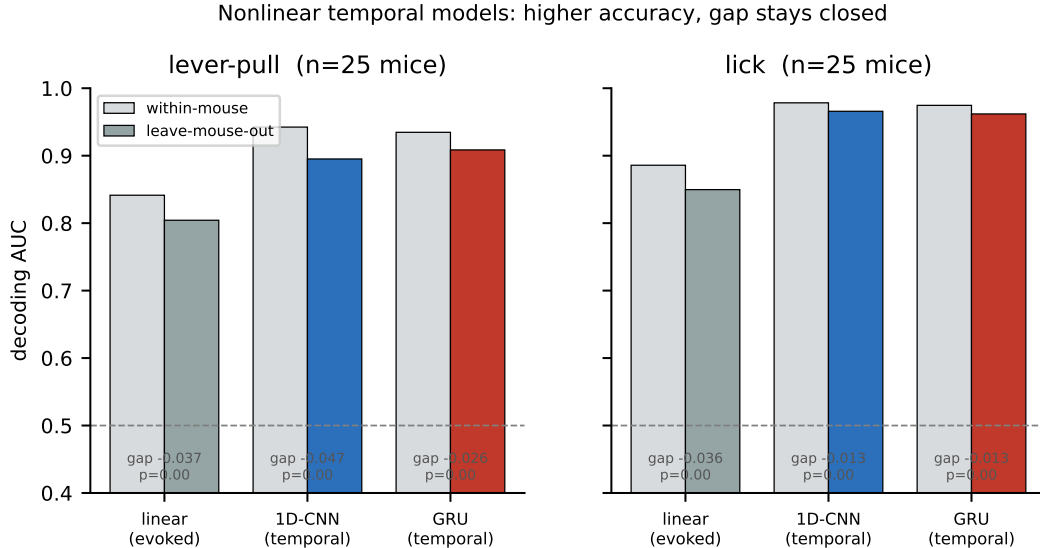


Figure 3: Within-mouse vs. LOMO AUC for the linear and two nonlinear temporal decoders. Capacity raises accuracy markedly but does not open a cross-animal gap.

are read from the dataset’s own ROI table and released as `parcel_names.json`.

5 DISCUSSION

On BraiDyn-BC, cortex-wide representations of all four task events generalize to unseen individuals within 0.03–0.04 AUC of within-animal decoding, and—our main contribution—for the two events we could test across blocks that small gap is stable across a 15-day protocol on both the within- and cross-animal axes and does not widen with model capacity. The shared component of the code is therefore invariant to individual identity, to session, and to whether the decoder is a 44-parameter linear map or a temporal network; and it includes the response dynamics, not only the collapsed evoked mean. That cortex-wide dynamics are conserved across individuals is not itself new (MacDowell & Buschman, 2020; MacDowell et al., 2024; Musall et al., 2019), and a recently published foundation model performs atlas-aligned cross-subject decoding on this cohort (Hosseini et al., 2026); our contribution is the interpretable, near-ceiling, and longitudinally- and capacity-controlled characterization those lines do not provide.

Limitations. We decode the four thresholdable task-event channels; continuous or graded behavioral variables are a natural next target, and are precisely where the cross-subject problem appears hardest (Hosseini et al., 2026). Our two characterization results are narrower than the cross-sectional baseline: the longitudinal (Section 4.2) and capacity (Section 4.3) analyses are run on lever-pull and lick only—the two events with enough trials per mouse per block to support an early/late split—so whether reward and tone are equally stable and equally capacity-invariant is untested. Our conclusions rest on LOMO-vs-within and across-vs-same-block comparisons rather than absolute AUC, and they hold for both the linear and the nonlinear decoders. The residual cross-animal gap is small but non-zero: we characterize rather than eliminate it. The longitudinal-novelty framing is, to the best of our literature search, unaddressed by prior work, but necessarily rests on absence of evidence. Finally, the drift literature we invoke as motivation (Rule et al., 2020) documents reorganization at the single-neuron level, whereas our signal is a mesoscale average over large populations within each parcel; such averaging may attenuate single-cell drift, so our stability result should be read as establishing that the parcel-level code a cross-animal decoder actually consumes is stable—not as evidence against single-neuron drift, which we cannot observe here.

6 CONCLUSION

A strict leave-one-mouse-out analysis in a shared 44-parcel atlas space shows that cortex-wide widefield representations of four behavioral events are read out across unseen mice within 0.03–0.04 AUC of the within-animal ceiling, and that on the two events we could test across blocks this small cross-subject gap is stable across a two-week protocol and does not widen with decoder capacity, with an independent-cohort replication. We release code and a streamed-feature pipeline that reproduces every number without downloading raw imaging.

Reproducibility statement

All data are open (BraiDyn-BC, DANDI:001425, CC-BY; Cardin–Higley, figshare 175317). We release the full analysis code and a streaming feature pipeline: every BraiDyn-BC table and figure is reproduced from the public parcellated traces without downloading raw widefield movies, and the committed result files (`braidyn_4targets.json`, `braidyn_drift.json`, `braidyn_nonlinear.json`) carry every reported value together with per-mouse breakdowns and the 44-parcel ROI table. The independent-cohort replication of Section 4.4 is the one analysis that is not streamed: it requires the Cardin–Higley parcel-trace archive to be fetched from figshare first, and re-running it emits `cardin_results.json`. All splits, permutation seeds, and model hyperparameters are specified in Section 3 and the released code.

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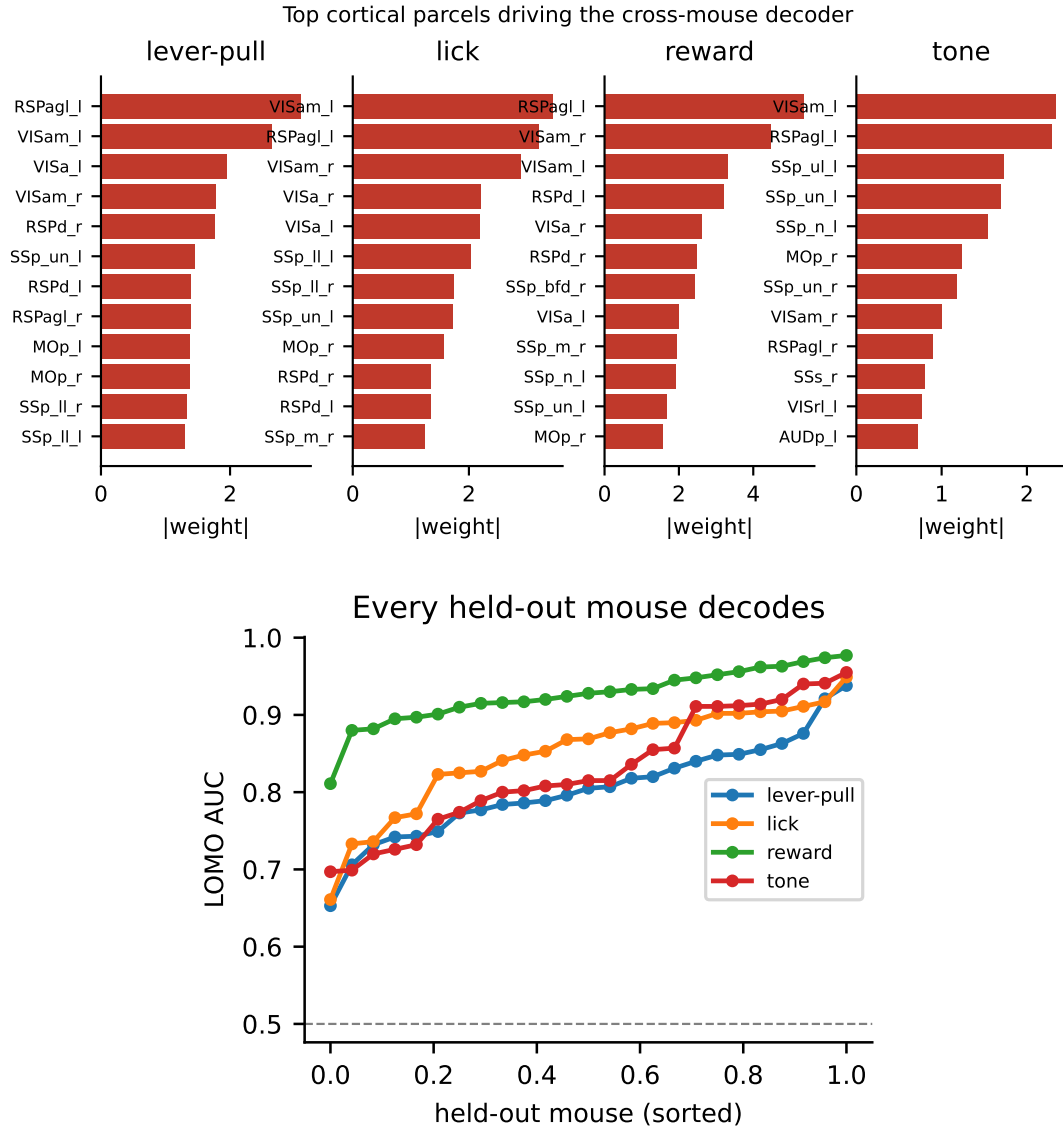


Figure 4: Top: top Allen-atlas parcels by decoder weight—a shared midline retrosplenial/visual-association backbone with target-specific somatosensory/motor modulation. Bottom: per-held-out-mouse LOMO AUC (sorted); every held-out mouse decodes above chance.